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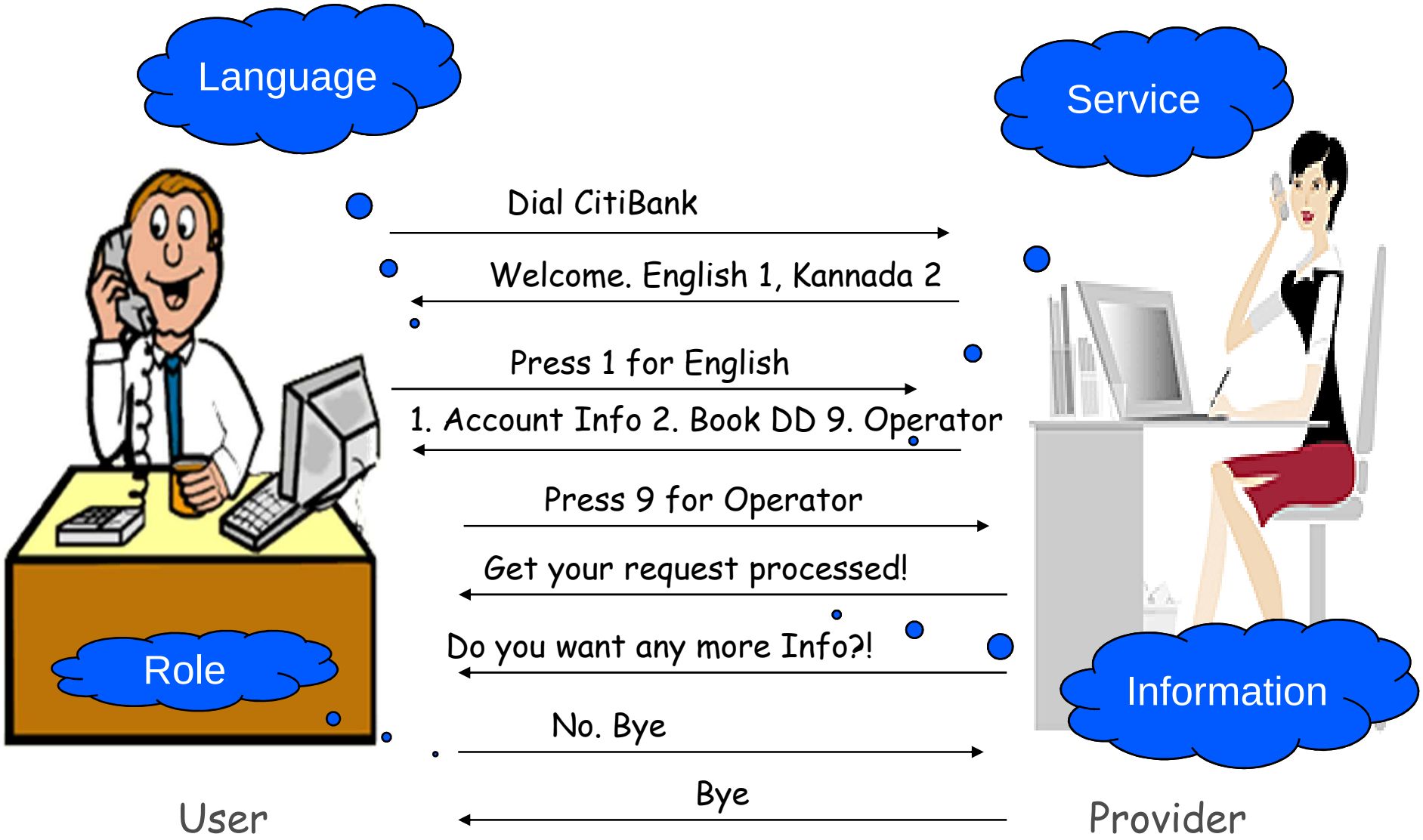
DICOM Network Model

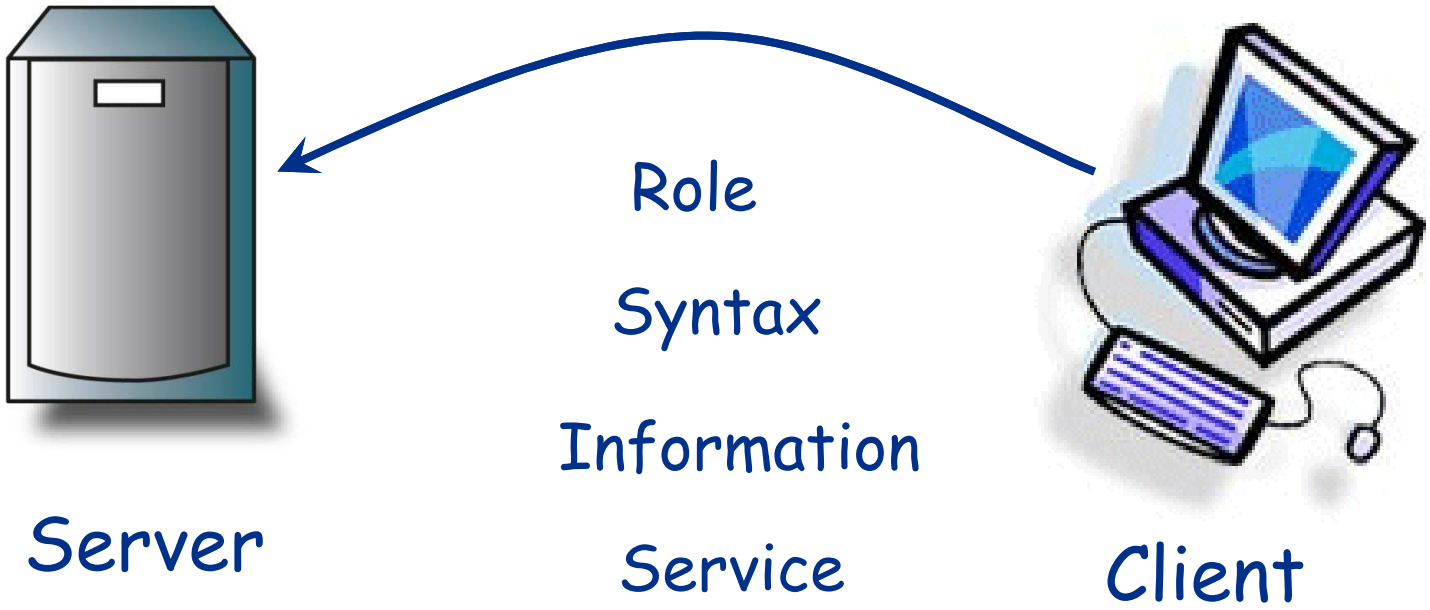
Agenda

- DICOM Network Model
- DICOM Association
- Presentation Context

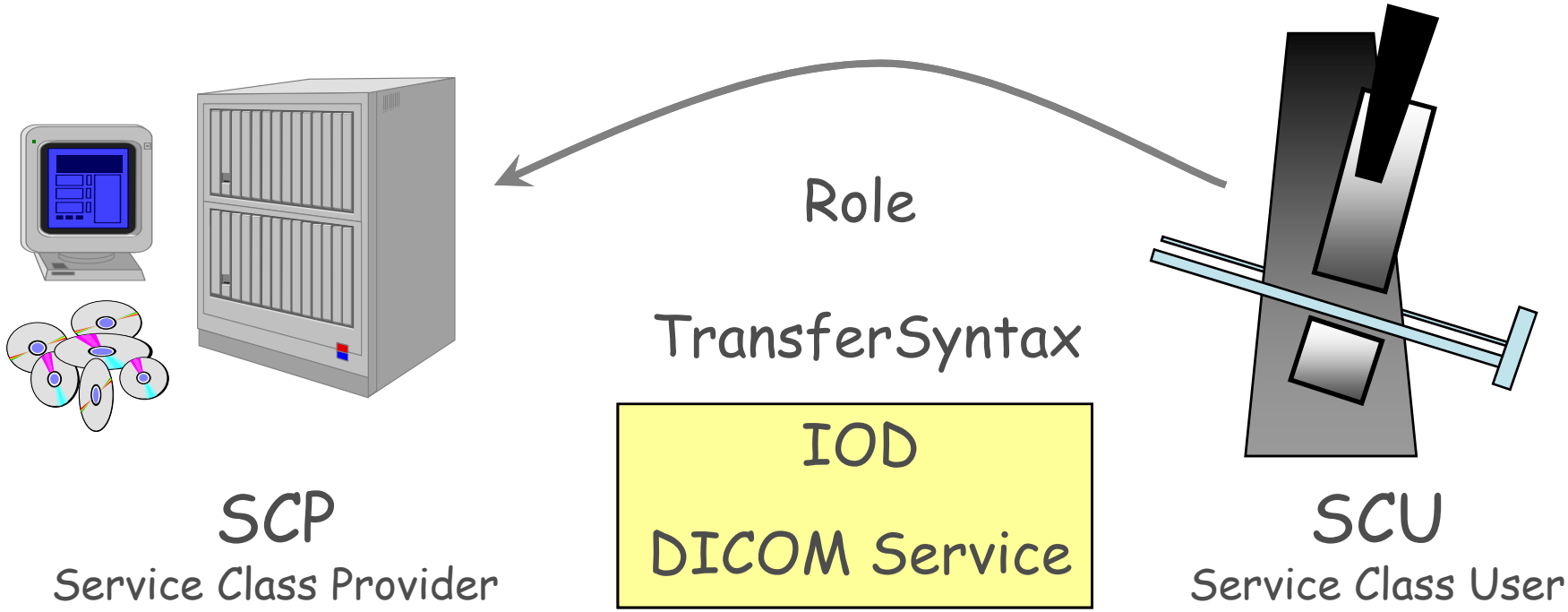
Communication



Client Server Model

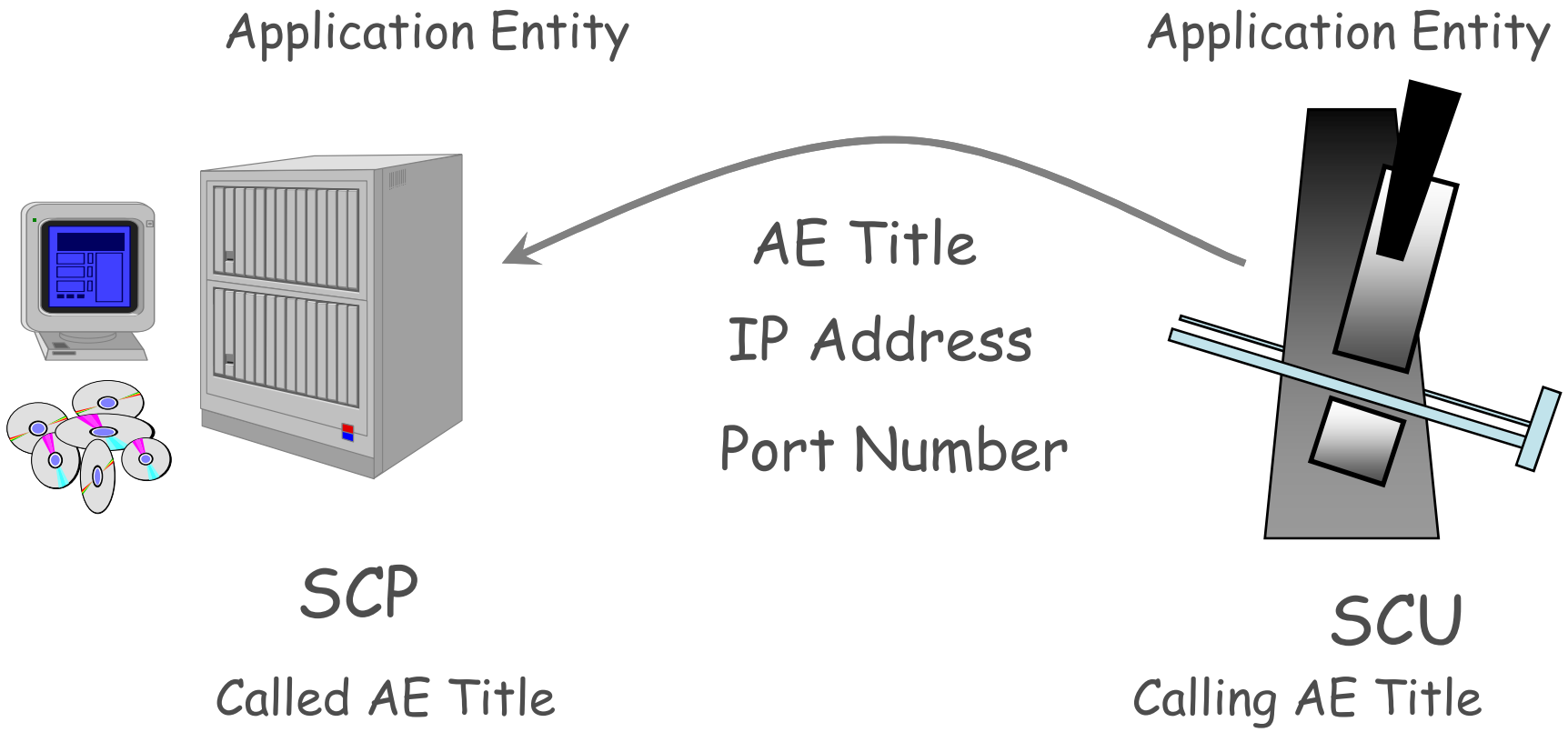


DICOM Network Model

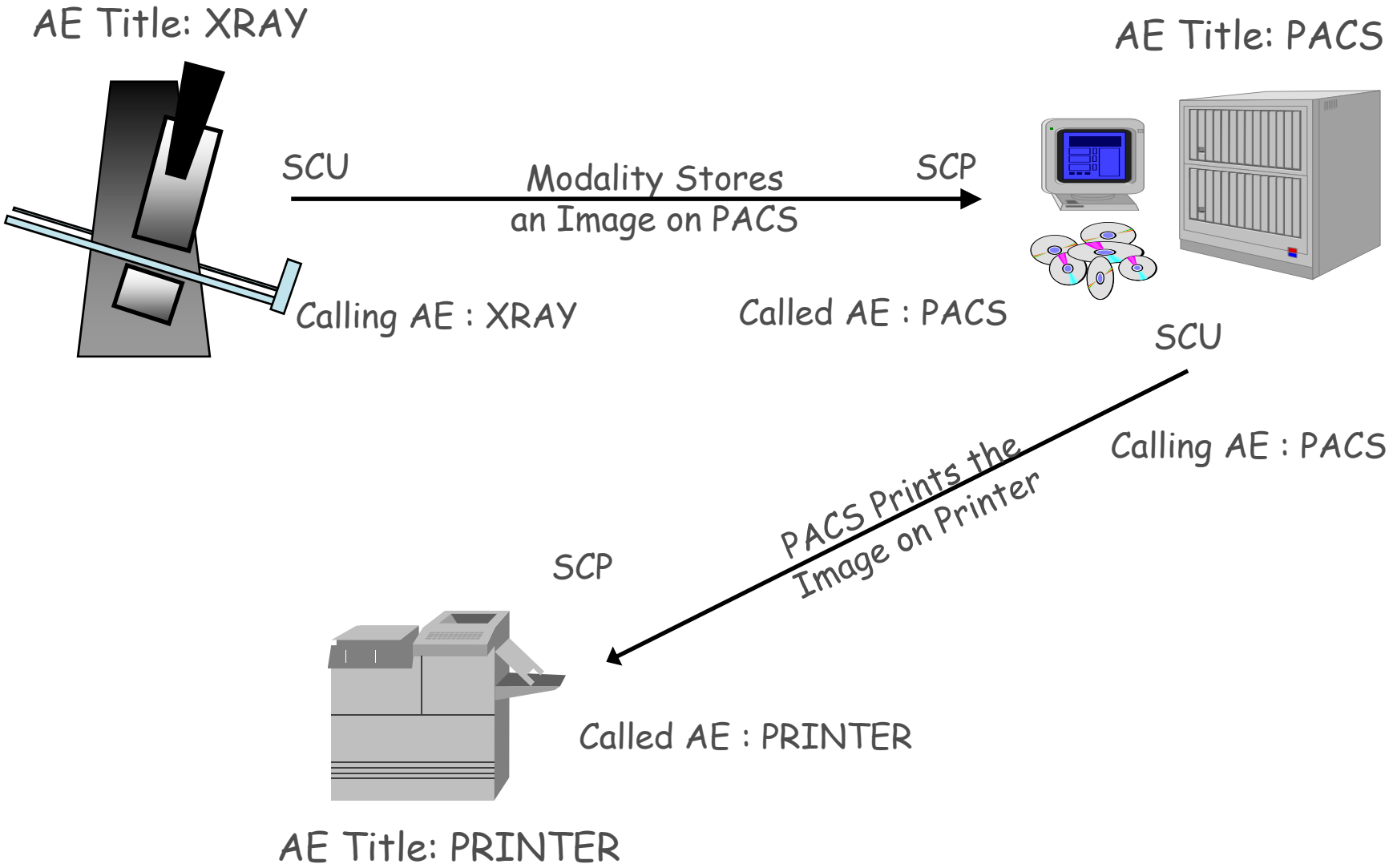


Service Object Pair (SOP)

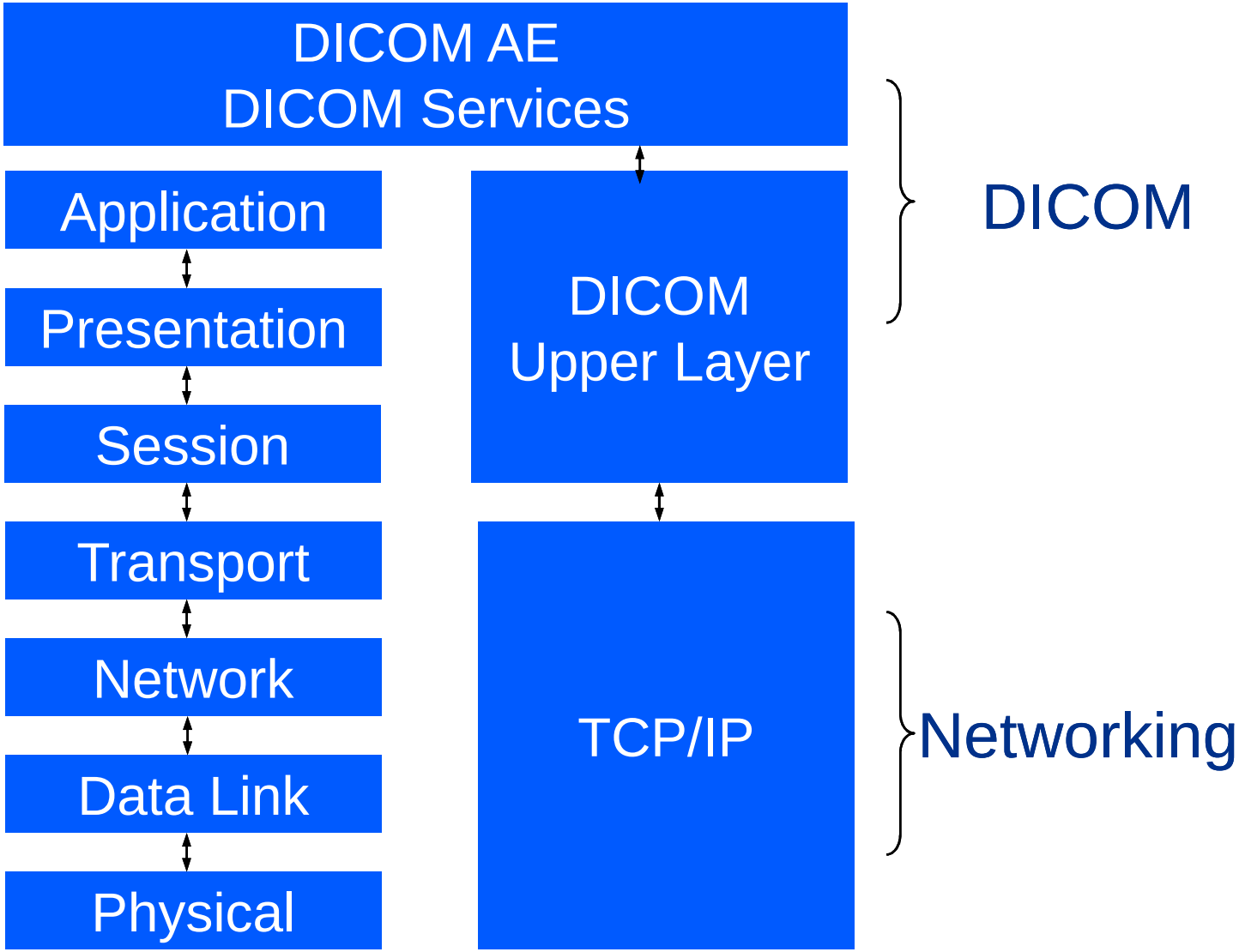
DICOM Communication



DICOM Network Model



DICOM Network Model



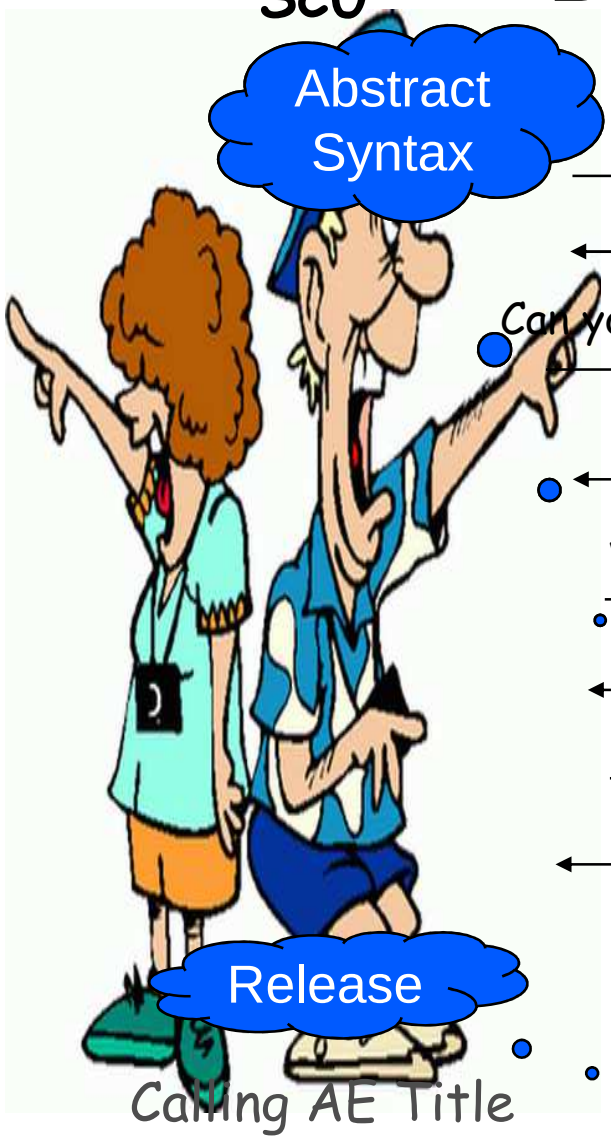
DICOM Network Model

DICOM Handshake

Known as

DICOM Association

DICOM Association



Hello!!!

Hello!!

Can you speak English/Spanish or Italian

English is Fine !!

We wanted to know the Directions

Sure I can help you with that

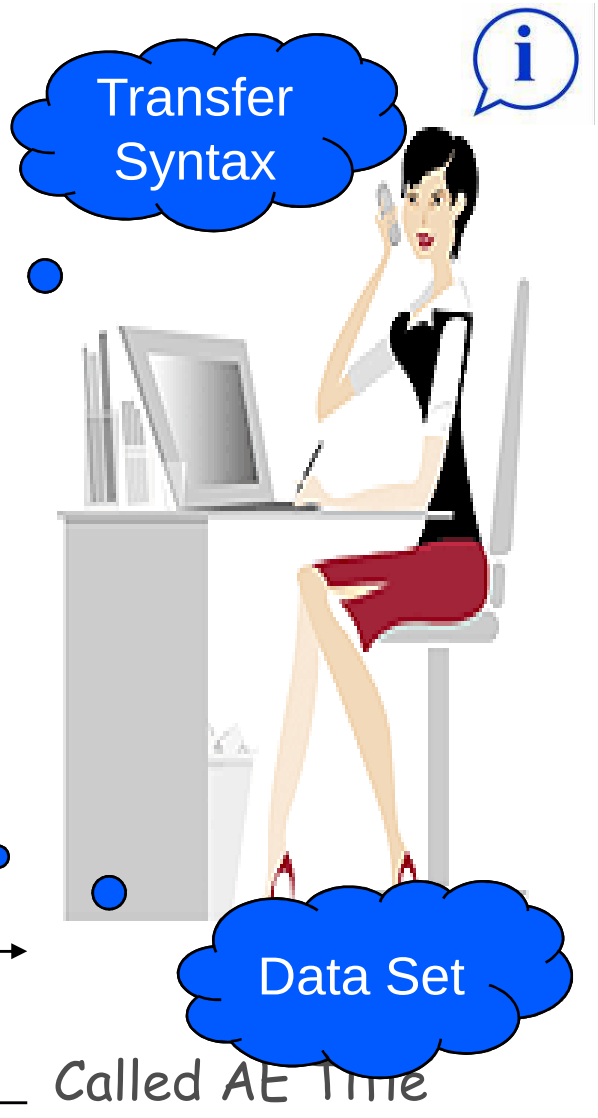
Provide details about where to go

Provides the Information

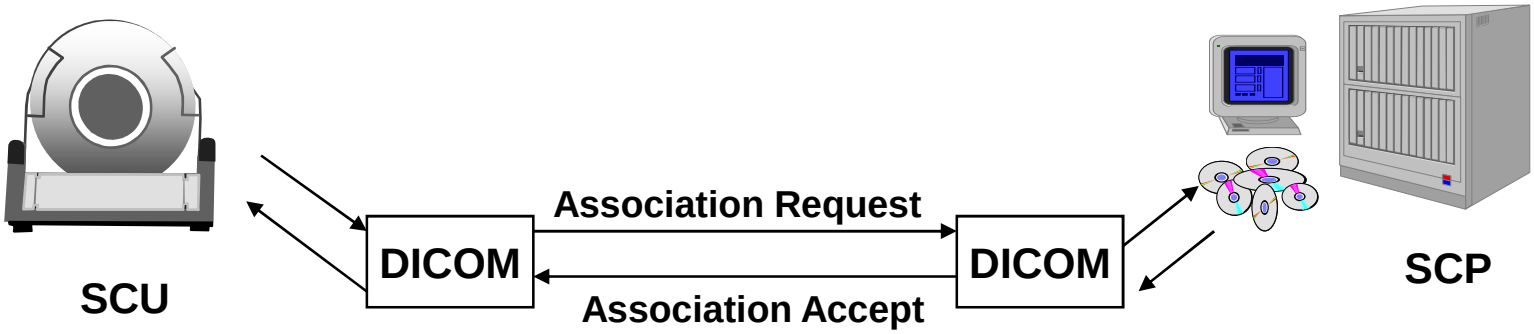
Thanks for the Information

Bye !!

Bye



DICOM Association



- DICOM Association
  - SCU Initiates
  - SCP Accepts

- Association Message Contains
  - Called AE Title
  - Calling AE Title
  - IP Address & Port Numbers
  - One or more Presentation Contexts

DICOM Association

Presentation Context = Abstract Syntax + Transfer Syntax

Where
Abstract Syntax is the SOP Class for the Service
Transfer Syntax is the Format of Data Encoding

e.g: Storage of an MR Image in Implicit VR Little Endian

Where:
Abstract Syntax = MR Image Storage
Transfer Syntax = Implicit VR Little Endian

Presentation Context

SCU Proposes

- One Abstract Syntax
- One or more Transfer Syntaxes for each Abstract Syntax

Presentation Context 1	Abs Syn MR Image Storage	TS: Implicit VR Little Endian	TS: Explicit VR Little Endian	TS: JPEG Lossless

Presentation Context 3	Abs Syn US Image Storage	TS: Implicit VR Little Endian	TS: Explicit VR Big Endian	TS: RLE Lossless

Presentation Context 5	Abs Syn Query for Patient	TS: Implicit VR Little Endian	TS: Explicit VR Big Endian

Presentation Context 7	Abs Syn Grayscale Print	TS: Implicit VR Little Endian	TS: Explicit VR Little Endian

Presentation Context

- SCP Accepts
  - One Abstract Syntax
  - One Transfer Syntax

- SCP can also Reject one or more Presentation Contexts

Presentation Context 1	Abs Syn MR Image Storage	TS: Implicit VR Little Endian	TS: Explicit VR Little Endian	TS: JPEG Lossless

~~Presentation Context 3~~	Abs Syn US Image Storage	TS: Implicit VR Little Endian	TS: Explicit VR Big Endian	TS: RLE Lossless

Presentation Context 5	Abs Syn Query for Patient	TS: Implicit VR Little Endian	TS: Explicit VR Big Endian

~~Presentation Context 7~~	Abs Syn Grayscale Print	TS: Implicit VR Little Endian	TS: Explicit VR Little Endian

Presentation Context

Presentation Context 1	Abs Syn MR Image Storage	TS: Implicit VR Little Endian	TS: Explicit VR Little Endian	TS: JPEG Lossless
Presentation Context 3	Abs Syn US Image Storage	TS: Implicit VR Little Endian	TS: Explicit VR Big Endian	TS: RLE Lossless
Presentation Context 5	Abs Syn Query for Patient	TS: Implicit VR Little Endian	TS: Explicit VR Big Endian	
Presentation Context 7	Abs Syn Grayscale Print	TS: Implicit VR Little Endian	TS: Explicit VR Little Endian	

Presentation Contexts Proposed

Presentation Context 1	Abs Syn MR Image Storage	TS: Explicit VR Big Endian
Presentation Context 5	Abs Syn Query for Patient	TS: Implicit VR Big Endian

Presentation Contexts Accepted

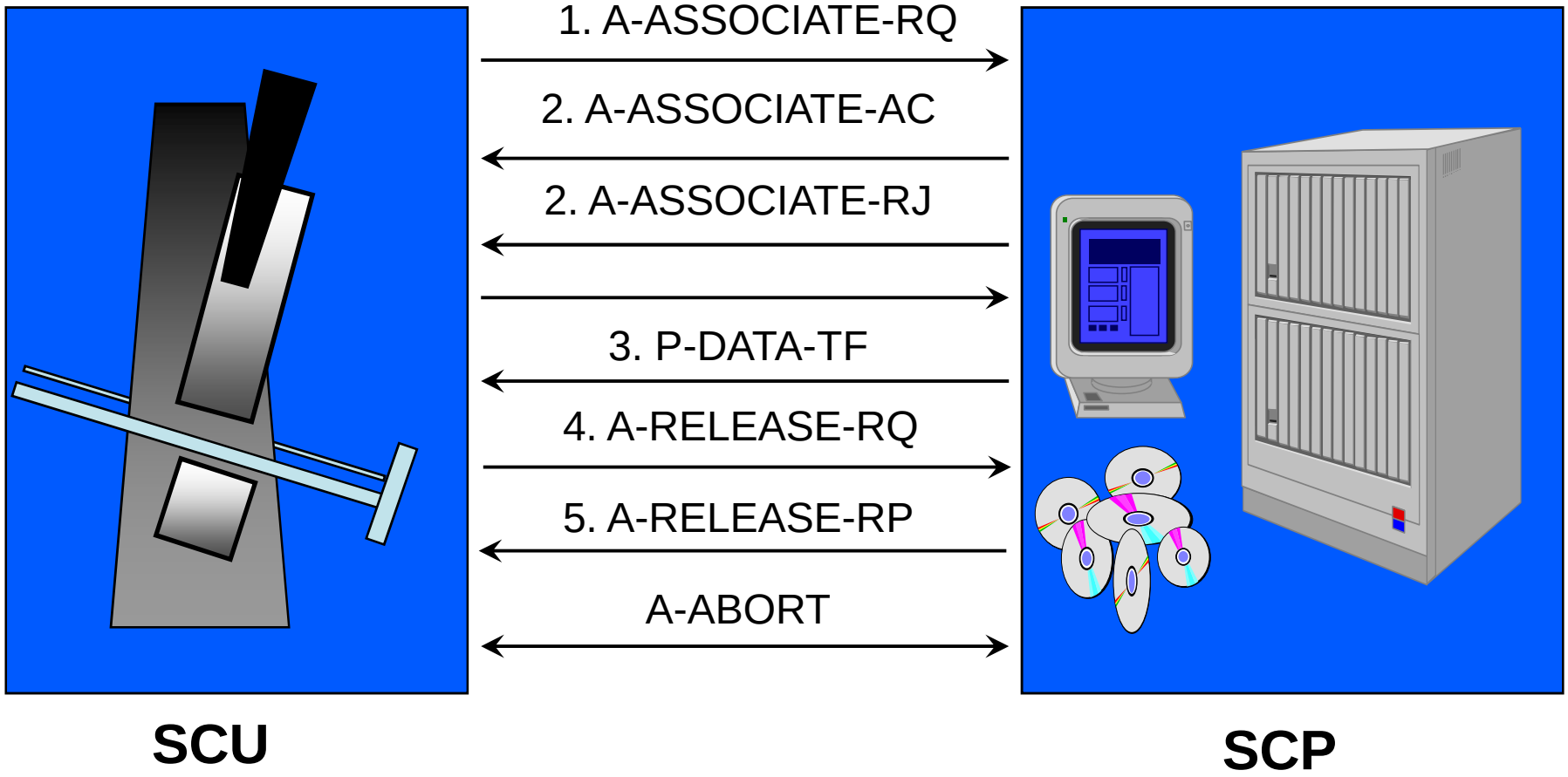
DICOM Upper Layer

- DICOM Messages are encoded in Protocol Data Units (PDU)

- Seven Types of DICOM PDUs

  – A-ASSOCIATE-RQ
  – A-ASSOCIATE-AC
  – A-ASSOCIATE-RJ
  – P-DATA-TF
  – A-RELEASE-RQ
  – A-RELEASE-RP
  – A-ABORT

DICOM Communication

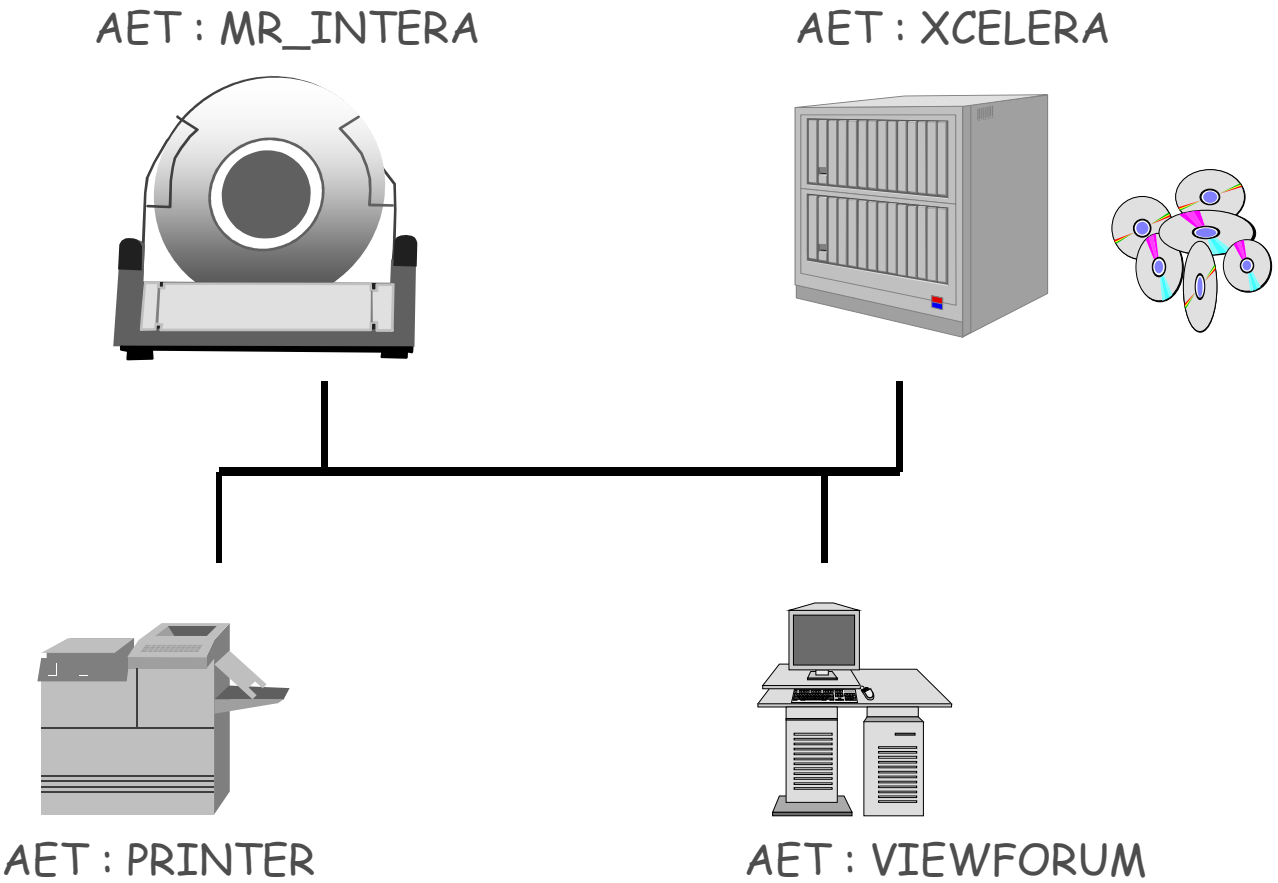


DICOM Network Model

- Demo
  - Service Class User/Service Class Provider
  - DICOM Association
  - Presentation Context

Exercise

- Identify the Various Transactions, Roles along with Called AE/Calling possible in the picture below



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sense and simplicity